Near-Square-Root Regret for Adversarially Replicable Experts

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Abstract

We study the full-information experts problem with adversarial replicability: two independent cost sequences from any fixed, time-varying product law must induce identical entire action paths when the learner uses the same random tape. We give a causal algorithm with worst-case expected regret $O_{n,\rho}(\sqrt{T}\log^{7/2}(2T))$ for every fixed number of experts n and replication parameter ρ. A two-expert construction combines a relay with a history-dependent switching threshold, dyadic evidence windows, and a rare compensating prediction rule. A tournament reduction uses modal output paths only in its analysis to preserve the product-law guarantee. The dependence on n and ρ is poor. The result concerns expected regret and does not establish a strict $O(\sqrt{T})$ bound, efficient finite-precision implementation, or a guarantee for adaptive input laws.

1 Introduction and model

Ahmadi, Bhandari and Blum [1] introduced adversarial replicability for online learning. In their experts model, an oblivious adversary fixes an arbitrary sequence of distributions over cost vectors. Two independent draws from those distributions, processed with the same internal random tape, must produce the same *entire* sequence of expert choices with high probability. Their Theorem 4.1 gives an experts regret bound $\widetilde{O}(T^{5/6}(\ln n)^{5/6}\rho^{-1/3})$. We obtain a square-root dependence on the horizon, up to logarithmic factors, when n and ρ are fixed. Our dependence on those two parameters is substantially worse. This is a comparison with the stated theorem, not a claim that its bound is the best currently available.

The construction has three parts. A two-expert rule changes action only when a public dyadic window supports the change. Its threshold grows with an input-only measure of profitable interval variation. A second rule, used on a small probability branch, compensates for a variation-dependent term in the first rule's expected regret. Finally, a binary tournament extends the result to finitely many experts. For the tournament's replication proof, we replace each child's random path by a modal path *in the analysis only*. This avoids assuming that the actual adaptive losses seen by the parent are independent across time.

The compensating rule uses Cover's classical backward fair-future betting construction [2, Section 3, equations (3.6)–(3.8)], extended from binary to $[-1,1]$ inputs by coordinatewise convexity. We include the short argument in Section 7; the fair-future formula and this immediate extension are not claimed as new contributions. The new bound retains logarithmic factors. It therefore does not resolve the literal alternative in Section 6 of [1]: a strict $O(\sqrt{T})$ upper bound versus an $\omega(\sqrt{T})$ lower bound at fixed replication probability.

Model and conventions. There are n experts and a known horizon T. At time t, the learner chooses $I_t \in [n]$, observes $c_t \in [0,1]^n$, and incurs $c_t(I_t)$. The choice is a measurable function of

$c_1, \dots, c_{t-1}$ and an input-independent tape R. For each deterministic input $c_{1:T}$, define

$$\mathrm{Reg}_A(c_{1:T}) = \mathbb{E}_R \sum_{t=1}^{T} c_t(I_t) - \min_{i\in[n]} \sum_{t=1}^{T} c_t(i).$$

The expectation includes every branch of the original tape. Regret is not truncated at zero and need not be nonnegative on an individual input. We do not require a seedwise or high-probability regret bound.

For every fixed sequence of distributions $D_1, \dots, D_T$ on $[0,1]^n$, let C, C' be independent draws from $P = \bigotimes_{t=1}^{T} D_t$, independent of R. The learner is *adversarially ρ-replicable* if

$$\mathbb{P}_{R,C,C'}\{I_t(C,R) = I_t(C',R) \text{ for all } t \in [T]\} \geq 1-\rho. \tag{1}$$

There is no identical-distribution assumption, and coordinates of a single cost vector may be dependent. Independence across times and oblivious choice of the laws are essential to the stated model.

All logarithms are natural unless the base is displayed. Put $z_+ = \max\{z, 0\}$ and $\mathrm{clip}(z) = \max\{-1, \min\{z, 1\}\}$. For two experts write $x_t = c_t(2) - c_t(1) \in [-1, 1]$ and let $A_t = +1$ select expert 1 and $A_t = -1$ select expert 2. Then

$$S_t = \sum_{i=1}^{t} x_i, \qquad G_T = \sum_{t=1}^{T} A_t x_t, \qquad \mathrm{Reg}_A = \frac{|S_T| - \mathbb{E}_R G_T}{2}. \tag{2}$$

This identity follows by subtracting the common cost $(c_t(1) + c_t(2))/2$ at each time. It imposes no complementary-cost restriction on the original input vectors.

2 Main guarantees and finite parameters

For $T \geq 2$ define

$$K = 2\lceil \log_2(T+1) \rceil + 2, \qquad J = 1 + \lfloor \log_2(T-1) \rfloor, \qquad \gamma = \sqrt{8 \ln(2T)}. \tag{3}$$

For an outer replication parameter $\rho \in (0,1)$ set

$$\beta = \rho/4, \qquad \lambda = \frac{\beta}{1-\beta}, \qquad r = \rho/2, \tag{4}$$

$$\alpha = \lambda/120, \qquad \epsilon = \frac{\lambda}{80(K+1)}, \qquad \eta = \sqrt{2T \ln(16T/r)}. \tag{5}$$

We use the explicit threshold

$$H(T,\rho) = \max\left\{ \frac{1}{\epsilon}, \quad \gamma\sqrt{(K+1)T}, \quad \frac{128 J \eta}{\alpha \epsilon^2 r} \right\}, \tag{6}$$

where all quantities on the right have the dependence in (3)–(5). Define

$$B(T,\rho) = 10 H(T,\rho) + \frac{\rho}{8}(\sqrt{T} + 1). \tag{7}$$

For each fixed ρ, this is $O_\rho(\sqrt{T}\log^{7/2}(2T))$: $J, K = O(\log(2T))$, $\epsilon^{-1} = O_\rho(K+1)$, and $\eta = O_\rho(\sqrt{T\log(2T)})$.

Theorem 2.1 (Two experts)**.** *For $T \geq 2$ and $\rho \in (0,1)$, the algorithm specified in Sections 4 and 7 has $\mathrm{Reg}_A(c_{1:T}) \leq B(T,\rho)$ on every deterministic $[0,1]^2$ cost sequence and satisfies (1). For $T = 1$, a shared uniform expert has exact replication and expected regret at most $1/2$.*

Theorem 2.2 (Fixed finite number of experts)**.** *For $n \geq 2$, let $N = 2^{\lceil \log_2 n \rceil}$. There is a causal full-information adversarially ρ-replicable learner with*

$$\sup_{c_{1:T} \in ([0,1]^n)^T} \mathrm{Reg}_A(c_{1:T}) \leq (\log_2 N) B(T, \rho/N^2), \qquad T \geq 2. \tag{8}$$

Consequently its regret is $O_{n,\rho}(\sqrt{T}\log^{7/2}(2T))$ for every fixed n and ρ. For $n = 1$ the sole expert suffices. For $T = 1$, a shared input-independent expert choice has exact replication and regret at most one.

The proof of Theorem 2.1 concludes in Section 7. The tournament in Section 8 proves Theorem 2.2.

3 An interval-variation functional

Set $S_0 = V_0 = 0$ and define

$$V_t = \max_{\mathcal{I}} \sum_{I \in \mathcal{I}} \left(\left| \sum_{i \in I} x_i \right| - \gamma\sqrt{|I|} \right)_+, \qquad Q = (V_T - |S_T|)_+. \tag{9}$$

Here $\mathcal{I}$ ranges over collections of pairwise disjoint nonempty integer intervals in $[1,t]$, including the empty collection. The maximum is finite. In particular $0 \leq V_t \leq t$.

Lemma 3.1. *The functional V_t is nondecreasing in t, coordinatewise convex, and 1-Lipschitz in each input coordinate. For $0 \leq s < t \leq T$,*

$$V_t \geq V_s + (|S_t - S_s| - \gamma\sqrt{t-s})_+, \tag{10}$$

$$V_t \leq |S_t| + Q + \gamma\sqrt{T}. \tag{11}$$

Proof. Append the interval $(s,t]$ to an optimal collection in $[1,s]$ to obtain (10). Retaining a collection proves monotonicity. For each fixed collection, a coordinate appears in at most one summand, whose positive-part absolute-value expression is convex and 1-Lipschitz in that coordinate. Maximizing preserves both properties. Applying (10) to the suffix $(t,T]$ yields

$$V_t \leq V_T - |S_T - S_t| + \gamma\sqrt{T} \leq |S_T| + Q - |S_T - S_t| + \gamma\sqrt{T} \leq |S_t| + Q + \gamma\sqrt{T}.$$

The case $t = T$ follows directly from $V_T \leq |S_T| + Q$. We will not assume that Q is convex. □

4 The supported relay

For the moment take positive parameters H, α, ϵ. Draw independent $\theta \sim \mathrm{Unif}[-1,1]$ and $U \sim \mathrm{Unif}[1,2]$, independent of the input. Put $d = U\epsilon$ and $h_t = H + \alpha V_t$. Start with $A_1 = +1$ if $\theta \leq 0$ and $A_1 = -1$ otherwise.

After observing x_t, a *public window* is $(t-L,t]$, where L is a power of two dividing t. Define flags

$$P_t^+ = \mathbf{1}\{S_t - S_{t-L} \geq dh_t \text{ for some public window}\},$$
$$P_t^- = \mathbf{1}\{S_t - S_{t-L} \leq -dh_t \text{ for some public window}\}.$$

For $t < T$, update by the following fixed rule:

1. If $A_t = -1$, $P_t^+ = 1$, and $S_t > h_t(\theta + d)$, set $A_{t+1} = +1$.
2. If $A_t = +1$, $P_t^- = 1$, and $S_t < h_t(\theta - d)$, set $A_{t+1} = -1$.
3. Otherwise keep $A_{t+1} = A_t$.

Windows are never restarted or truncated at a switch. Ties use exactly the comparisons displayed above. For the regret proof only, apply the same update virtually after time T; it changes no played action.

Conditional on U and a deterministic input, there is a scalar endpoint $a_t \in [-1, 1]$ such that the action after observation t is $+1$ precisely for $\theta < a_t$, apart from null ties. Initially $a_0 = 0$. If $P_t^+ = 1$, replace a by $\max\{a, \mathrm{clip}(S_t/h_t - d)\}$; if $P_t^- = 1$, replace it by $\min\{a, \mathrm{clip}(S_t/h_t + d)\}$. These updates commute when both flags hold, since their thresholds are ordered. Therefore

$$\mathbb{E}_\theta[G_T \mid U] = \sum_{t=1}^{T} a_{t-1} x_t. \tag{12}$$

This identity uses the original uniform center, without conditioning on agreement of two input streams.

5 Reward of the supported relay

This section considers a fixed deterministic input $x_1, \ldots, x_T \in [-1, 1]$, with $T \geq 2$. All expectations in the reward argument are over the learner's original random tape. In particular, none of the inequalities below is a regret guarantee for each individual random tape.

Recall that $S_t = \sum_{i=1}^{t} x_i$, $S_0 = 0$, $\gamma = \sqrt{8 \log(2T)}$, and

$$V_t = \max_{\mathcal{I}} \sum_{I \in \mathcal{I}} \left(\left| \sum_{i \in I} x_i \right| - \gamma \sqrt{|I|} \right)_+, \qquad V_0 = 0, \qquad Q = (V_T - |S_T|)_+ ,$$

where $\mathcal{I}$ ranges over collections of pairwise disjoint, nonempty integer intervals contained in $[1, t]$, including the empty collection. Let $h_t = H + \alpha V_t$. The width h_t is a nondecreasing function of the input alone. We condition throughout most of the proof on a fixed value $U \in [1, 2]$, and write $d = U\epsilon$. The remaining random center θ is uniform on $[-1, 1]$.

For clarity, the public support predicates after observation t are

$$\mathcal{P}_t : \quad \exists L \text{ a power of two with } L \mid t, \quad S_t - S_{t-L} \geq d h_t,$$

and the analogous predicate $\mathcal{N}_t$ with $S_t - S_{t-L} \leq -d h_t$. These predicates do not depend on θ. All such windows are allowed, including windows that precede a previous action change. The relay starts with $A_1 = +1$ if and only if $\theta \leq 0$. After observation t, a current action -1 changes to $+1$ when $\mathcal{P}_t$ holds and $S_t > h_t(\theta + d)$; a current action $+1$ changes to -1 when $\mathcal{N}_t$ holds and $S_t < h_t(\theta - d)$. Otherwise the action is retained. We also define the update after $t = T$ for the algebra below; the resulting A_{T+1} is never played.

Set

$$K = 2\lceil \log_2(T + 1) \rceil + 2.$$

Any integer increment interval $(s, t]$ can be partitioned into at most K aligned dyadic intervals. Indeed, its maximal contained aligned dyadic intervals partition it, and at most two of each length occur. Every piece of such a partition is one of the public windows at its right endpoint.

5.1 The scalar state and two elementary path inequalities

Lemma 5.1 (Nested centers)**.** *For fixed U and fixed input, there are scalar states $a_0, \dots, a_T \in [-1, 1]$, with $a_0 = 0$, such that the centers whose action after observation t is $+1$ form $[-1, a_t]$, up to boundary points of zero measure. Consequently, if*

$$g_u = \mathbb{E}_\theta \left[\sum_{t=1}^{u} A_t x_t \,\middle|\, U \right], \qquad \Delta a_t = a_t - a_{t-1},$$

then

$$g_u = \sum_{t=1}^{u} a_{t-1} x_t = a_u S_u - \sum_{t=1}^{u} S_t \Delta a_t. \tag{13}$$

At every actual up update, meaning $\Delta a_t > 0$,

$$S_t \geq h_t(a_t + d), \tag{14}$$

with equality unless $a_t = 1$. At an actual down update, $S_t \leq h_t(a_t - d)$, with equality unless $a_t = -1$.

Proof. Write $\mathrm{clip}(z) = \max\{-1, \min\{1, z\}\}$, and let $l_t = \mathrm{clip}(S_t/h_t - d)$ and $u_t = \mathrm{clip}(S_t/h_t + d)$. If only positive support holds, the new scalar is $\max\{a_{t-1}, l_t\}$; if only negative support holds, it is $\min\{a_{t-1}, u_t\}$. When both supports hold, the new scalar is the projection of a_{t-1} onto $[l_t, u_t]$. The latter interval is nonempty, and the two one-sided operations commute. If neither support holds, the scalar is unchanged. This proves the interval representation by induction, starting from $a_0 = 0$. Its signed action mean is exactly a_t. There are only finitely many center boundary values for this fixed input, so their treatment has no effect on the expectation. Summation by parts gives (13). At an actual up update, $a_t = \mathrm{clip}(S_t/h_t - d) > a_{t-1}$, which proves (14) and its equality statement. The down case is symmetric. □

Lemma 5.2 (Prefix localization)**.** *For every $0 \leq u \leq T$,*

$$V_u \leq |S_u| + Q + \gamma\sqrt{T}. \tag{15}$$

Proof. For $u < T$, append the suffix interval $(u, T]$ to a maximizing collection for V_u. This gives

$$V_T \geq V_u + \left(|S_T - S_u| - \gamma\sqrt{T - u} \right)_+ \geq V_u + |S_T - S_u| - \gamma\sqrt{T}.$$

Since $V_T \leq |S_T| + Q$ and $|S_T| - |S_T - S_u| \leq |S_u|$, the claim follows. For $u = T$ it is immediate from the definition of Q. □

For the generic estimates that follow, assume

$$0 < \alpha \leq \frac{1}{8}, \qquad H \geq \frac{1}{\epsilon}, \qquad (K+1)d \leq 1. \tag{16}$$

Because $d = U\epsilon \geq \epsilon$ and $K \geq 4$, these assumptions imply

$$d \leq \frac{1}{5}, \qquad M := 1 + d \leq \frac{3}{2}, \qquad D := Kd + \frac{1}{H} \leq (K+1)d \leq 1.$$

We use the abbreviation $\ell = d + \frac{3}{2}D$.

5.2 Delayed crossings and scalar jumps

Lemma 5.3 (Delay and jump bound)**.** *Suppose $r < t$ is the preceding actual scalar update before time t, or $r = 0$ if there is none, and put $a = a_r$. Thus a is fixed after r through just before t. If $a < 1$, then*

$$S_t - (a+d)h_t \le Dh_t + M(h_t - h_r).$$

If $a > -1$, then

$$S_t - (a-d)h_t \ge -Dh_t - M(h_t - h_r).$$

These bounds also hold when there is no actual update at time t. At every actual update, with r denoting its preceding actual update,

$$|\Delta a_t| \le D + M\frac{h_t - h_r}{h_t}. \tag{A}$$

Proof. Consider the upper bound, for which $a < 1$. Immediately after r,

$$S_r \le (a+d)h_r.$$

This holds initially. After an up update it is equality, because $a < 1$ rules out upper saturation. After a down update it follows from $S_r \le (a-d)h_r$. Let $b_s = (a+d)h_s$ and $B = \max_{r \le s \le t} b_s$.

If $S_t \le B$, the bound $S_t \le B + Kdh_t + 1$ is automatic. Otherwise, let $s < t$ be the last integer with $s \ge r$ and $S_s \le B$; such an s exists by the preceding endpoint inequality. Every integer v with $s < v < t$ has $S_v > B \ge b_v$. Partition $(s, t-1]$, if nonempty, into at most K aligned dyadic intervals. At the endpoint v of any such piece, its increment must be strictly less than dh_v. If not, positive support would hold there, and $S_v > (a+d)h_v$ with $a < 1$ would strictly increase the scalar before t, contradicting the choice of r. Each piece thus contributes less than dh_t, and the last increment x_t is at most one. It follows that

$$S_t \le B + Kdh_t + 1.$$

Since $|a+d| \le 1+d = M$ and h is nondecreasing,

$$B - (a+d)h_t \le M(h_t - h_r).$$

Also $1 \le h_t/H$, so the upper bound follows from the definition of D. The lower bound is proved in the same way using the minimum of $(a-d)h_s$ and negative support. Its starting inequality is valid whenever $a > -1$, by the symmetric endpoint argument.

At an actual up update, clipping can only reduce the movement, so

$$\Delta a_t \le S_t/h_t - d - a.$$

The upper delay bound gives (A); a down update uses the lower bound. The argument used only the absence of intermediate actual updates, not the existence of an update at t, which proves the stated final-tail version as well. Allowing old support windows causes no difficulty: all the internal windows used in the partition remain allowed. □

5.3 Weighted variation and saturation losses

For $u \le T$ define

$$W_u = \sum_{t=1}^{u} h_t \left|\Delta a_t\right|.$$

For $c > 0$, let $\mathrm{TV}^c(S_{[0,u]})$ be the supremum of $\sum_j \left(\left|S_{v_j} - S_{r_j}\right| - c\right)_+$ over pairwise disjoint increment intervals $(r_j, v_j] \subseteq (0, u]$.

Lemma 5.4 (Weighted variation)**.** *For every $u \le T$,*

$$\begin{aligned} W_u &\le \mathrm{TV}^{2dH}(S_{[0,u]}) + (h_u - H) + dH \\ &\le (1+\alpha)V_u + \frac{\gamma^2 T}{8dH} + dH. \end{aligned} \tag{B}$$

Proof. If there are no actual updates, the claim is immediate. Otherwise, partition the nonzero scalar changes through u into maximal runs with constant sign. Include intervening zero changes in each run interval. Suppose an up run ends at s and the preceding down run ends at r. Summation by parts, using $|a_t| \le 1$, gives

$$\begin{aligned} \sum_{t=r+1}^{s} h_t \Delta a_t &= h_s a_s - h_r a_r - \sum_{t=r+1}^{s} a_{t-1}(h_t - h_{t-1}) \\ &\le h_s a_s - h_r a_r + (h_s - h_r). \end{aligned}$$

The actual endpoint inequalities, including saturated endpoints, imply $h_s a_s \le S_s - dh_s$ and $h_r a_r \ge S_r + dh_r$. Hence this run's weighted variation is at most

$$S_s - S_r - d(h_s + h_r) + (h_s - h_r) \le (|S_s - S_r| - 2dH)_+ + (h_s - h_r).$$

For a down run the identical bound follows after reversing signs.

The first run starts at $r = 0$ with $S_0 = a_0 = 0$, rather than at an opposite-direction endpoint. If it is an up run, its bound before positive-part truncation is $S_s - dh_s + (h_s - H)$. The elementary inequality

$$S_s - dh_s \le (|S_s| - 2dH)_+ + dH$$

shows that one additional dH suffices. The first down run is symmetric. The run intervals are disjoint in their increments, and their total width increase is no greater than $h_u - H$. Summing proves the first line of (B). In particular, saturation has not been replaced by an equality at any run endpoint.

For any $c > 0$ and any interval of length L, $\gamma\sqrt{L} \le c + \gamma^2 L/(4c)$. Therefore

$$(|S_v - S_r| - c)_+ \le \big(|S_v - S_r| - \gamma\sqrt{v-r}\big)_+ + \frac{\gamma^2(v-r)}{4c}.$$

The first terms summed over a disjoint interval collection are bounded by V_u, and the lengths sum to at most T. Take $c = 2dH$ and use $h_u - H = \alpha V_u$ to obtain the second line of (B). □

Lemma 5.5 (Energy accounting)**.** *For every $u \le T$,*

$$g_u \ge a_u S_u - \frac{h_u}{2} - \ell W_u - 3M(h_u - H). \tag{C}$$

Proof. At each actual update, define e_t by

$$S_t \Delta a_t = h_t a_t \Delta a_t + dh_t\,|\Delta a_t| + e_t.$$

At an unsaturated new value, $e_t = 0$. At an up saturation $a_t = 1$,

$$e_t = \big(S_t - h_t(1+d)\big)\Delta a_t \ge 0.$$

If r is the preceding actual update, the delay bound gives

$$S_t - h_t(1+d) \le h_t(D - \Delta a_t) + M(h_t - h_r) \le Dh_t + M(h_t - h_r).$$

Using $|\Delta a_t| \le 2$, this bounds e_t by $Dh_t\,|\Delta a_t| + 2M(h_t - h_r)$. The same bound holds at a down saturation. The intervals between consecutive actual updates are disjoint, so

$$\sum_{t \le u} e_t \le DW_u + 2M(h_u - H).$$

Similarly, multiply (A) by $h_t\,|\Delta a_t|/2$ and sum to obtain

$$\frac{1}{2}\sum_{t \le u} h_t(\Delta a_t)^2 \le \frac{D}{2}W_u + M(h_u - H).$$

Here any final interval with no scalar update can only increase the right-hand side through h_u.

The exact quadratic summation-by-parts identity is

$$\sum_{t \le u} h_t a_t \Delta a_t = \frac{1}{2}h_u a_u^2 + \frac{1}{2}\sum_{t \le u} h_t(\Delta a_t)^2 - \frac{1}{2}\sum_{t \le u} a_{t-1}^2(h_t - h_{t-1}).$$

The final term is nonpositive, since h is nondecreasing. Substitute this identity and the bounds above into (13), discard the favorable width term in the reward, and use $a_u^2 \le 1$. The result is (C), with $\ell = d + 3D/2$. □

5.4 Terminal localization and finite constants

Lemma 5.6 (A suitable terminal prefix)**.** *There exists $u \in \{0, \dots, T\}$, allowed to depend on U and the deterministic input, for which $|S_u| \le 4h_u$ and*

$$g_T \ge |S_T| - 9h_u - \ell W_u - 3M(h_u - H). \tag{D}$$

For this u,

$$V_u \le \frac{Q + \gamma\sqrt{T} + 4H}{1 - 4\alpha} \le 2\big(Q + \gamma\sqrt{T} + 4H\big). \tag{E}$$

Proof. First note that at any actual update t, $|S_t| \le 4h_t$. For an up update, the upper delay bound gives

$$S_t \le (1 + d + D + M)h_t < 4h_t,$$

where we used $d \le 1/5$, $D \le 1$, and $M \le 3/2$. The other side follows from $S_t \ge h_t(a_t + d) \ge -h_t$. A down update is symmetric.

If $a_T \in (-1, 1)$, choose $u = T$. If T is an actual update, the preceding bound applies. Otherwise, both one-sided delay bounds apply to the final interval of constant scalar state, and again yield $|S_T| \le (1 + d + D + M)h_T < 4h_T$. If a_T is saturated but $a_T S_T < 0$, also choose $u = T$. An actual update to a saturated value has S_T of the same sign, so in this case there is a final interval with no actual update. When $a_T = 1$ only the lower delay bound is needed, and when $a_T = -1$ only the upper bound is needed. Together with the assumed wrong sign, it gives $|S_T| \le 4h_T$. In both of these cases,

$$|S_T| - a_T S_T \le 2\,|S_T| \le 8h_T,$$

and (C) yields (D), since $8h_T + h_T/2 \le 9h_T$.

It remains to treat $a_T \in \{-1, 1\}$ with $a_T S_T \ge 0$. Let u be the last actual scalar update. Such an update exists because $a_0 = 0$. The actual-update bound above gives $|S_u| \le 4h_u$. After u, the scalar is constantly a_T, so

$$g_T = g_u + a_T(S_T - S_u).$$

Apply (C) at u and use $a_u = a_T$ and $a_T S_T = |S_T|$; this proves the stronger bound with $h_u/2$ in place of $9h_u$, and hence (D). This case includes $S_T = 0$. The virtual update at T, if it is the last actual update, is harmless: the displayed reward decomposition then has an empty final segment.

Finally, apply Lemma 5.2 at this u and use $|S_u| \le 4(H + \alpha V_u)$. Rearranging gives the first inequality in (E). The second uses $\alpha \le 1/8$. □

Proposition 5.7 (Generic reward estimate)**.** *Under* (16)*, for each fixed* $U \in [1, 2]$*,*

$$\begin{aligned} g_T \ge\ & |S_T| - (30\alpha + 4\ell)(Q + \gamma\sqrt{T}) \\ & - \big[9 + \ell d + 120\alpha + 16\ell\big] H - \frac{\ell\gamma^2 T}{8dH}. \end{aligned} \tag{F}$$

Proof. Substitute (B) into (D), and use $h_u = H + \alpha V_u$. With

$$b = 9\alpha + 3M\alpha + \ell(1 + \alpha),$$

the result is

$$g_T \ge |S_T| - (9 + \ell d)H - bV_u - \frac{\ell\gamma^2 T}{8dH}.$$

Since $M \le 3/2$ and $\alpha \le 1/8$,

$$2b \le 27\alpha + \frac{9}{4}\ell \le 30\alpha + 4\ell, \qquad 8b \le 108\alpha + 9\ell \le 120\alpha + 16\ell.$$

Apply (E) and these two conservative coefficient bounds to obtain (F). □

Lemma 5.8 (Good-branch reward)**.** *Let* $\lambda \in (0, 1]$*, and choose*

$$\alpha = \frac{\lambda}{120}, \qquad \epsilon = \frac{\lambda}{80(K+1)}, \qquad H \ge \max\left\{\frac{1}{\epsilon}, \gamma\sqrt{(K+1)T}\right\}.$$

The specified relay satisfies, for every deterministic input in $[-1, 1]^T$*,*

$$\mathbb{E}_{\theta,U}\left[\sum_{t=1}^{T} A_t x_t\right] \ge |S_T| - \lambda Q - 20H. \tag{G}$$

In fact, the same bound holds conditional on every fixed $U \in [1, 2]$*.*

Proof. For every such U, the generic assumptions hold. Moreover,

$$\ell = d + \frac{3}{2}D \le \frac{5}{2}(K+1)d \le 5(K+1)\epsilon = \frac{\lambda}{16}.$$

Consequently,

$$30\alpha + 4\ell \le \frac{\lambda}{2}, \qquad 9 + \ell d + 120\alpha + 16\ell \le 9 + \frac{\lambda}{80} + 2\lambda < 12.$$

The assumed lower bound on H implies $\gamma\sqrt{T} \le H$ and

$$\frac{\ell\gamma^2 T}{8dH} \le \frac{5(K+1)\gamma^2 T}{16H} \le \frac{5}{16}H.$$

Thus (F) gives the stronger estimate

$$g_T \ge |S_T| - \frac{\lambda}{2}Q - \left(12 + \frac{1}{2} + \frac{5}{16}\right)H.$$

Use $Q \ge 0$ to obtain (G). The estimates are uniform in U, so averaging over its original independent distribution preserves the bound. □

Remark 5.9. The last-update choice in Lemma 5.6 is essential: it avoids charging a large terminal variation after a long final segment on which the scalar action is already saturated with the correct sign. All scalar reward identities above average over the original random center. They impose neither a seedwise regret guarantee nor nonnegative regret on any branch of the eventual learner.

6 Whole-path replication of the supported relay

The widening statistic V_t and the support windows depend only on the observed costs, not on the relay's actions. This permits a comparison with deterministic quantities associated with the input law. In particular, no part of the following argument conditions on agreement of earlier actions.

Lemma 6.1 (Replication of the supported relay)**.** *Let $T \geq 2$, $0 < \alpha, \epsilon \leq 1$, and $0 < r < 1$. Set*

$$\eta = \sqrt{2T \ln(16T/r)}, \qquad J = 1 + \lfloor \log_2(T-1) \rfloor. \tag{17}$$

If

$$H \geq \frac{128 J \eta}{\alpha \epsilon^2 r}, \tag{18}$$

then the supported relay with $h_t = H + \alpha V_t$, shared independent $\theta \sim \mathrm{Unif}[-1,1]$ and $U \sim \mathrm{Unif}[1,2]$, and $d = U\epsilon$, is r-replicable. More precisely, for every fixed product law on $[-1,1]^T$, two independent input streams using the same complete random tape produce identical played-action paths $(A_1, \ldots, A_T)$ with probability at least $1 - r$. The law may vary arbitrarily across time and is independent of the tape. The conclusion allows any common tape-measurable initialization of A_1, including the initialization used above.

Proof. Fix such an input law. Write

$$\bar{S}_t = \mathbb{E} S_t, \qquad D_t = \mathbb{E} V_t, \qquad \bar{h}_t = H + \alpha D_t.$$

These are deterministic functions of the fixed law. A relevant monitoring window is $I = (t-L, t]$ with $1 \leq t < T$, L a power of two, and $L \mid t$. Let $\mu_I = \bar{S}_t - \bar{S}_{t-L}$. The two streams will be denoted by superscripts $1, 2$ when necessary.

First, for an arbitrary $\eta > 0$, there is an event $\mathcal{E}$ depending only on the two input streams such that

$$\mathbb{P}(\mathcal{E}^c) \leq 8T \exp\left(-\frac{\eta^2}{2T}\right), \tag{19}$$

and, on $\mathcal{E}$, for every prefix and both streams,

$$\left|S_t - \bar{S}_t\right| \leq \eta, \qquad |V_t - D_t| \leq \eta, \qquad \left|h_t - \bar{h}_t\right| \leq \alpha\eta. \tag{20}$$

Indeed, Hoeffding's inequality gives the stated tail bound for S_t. For a fixed interval collection defining V_t, a coordinate occurs in at most one summand, and changing that coordinate changes the summand by at most the magnitude of the change. Taking a maximum preserves this coordinatewise 1-Lipschitz bound. Since each input coordinate lies in $[-1,1]$, the bounded-differences inequality gives the same tail bound for V_t. A union bound over the two statistics, both streams, and at most T prefixes proves (19). Independence across time is used here; identical distribution is not required.

Appending I to an optimal interval collection in $[1, t-L]$ gives

$$V_t \geq V_{t-L} + \left(|S_t - S_{t-L}| - \gamma\sqrt{L}\right)_+ .$$

Taking expectations and using Jensen's inequality yields

$$D_t - D_{t-L} \geq |\mu_I| - \gamma\sqrt{L}. \tag{21}$$

Also D_t and $\bar{h}_t$ are nondecreasing in t.

For the moment assume

$$\epsilon H \geq 4(1+\alpha\epsilon)\eta, \qquad \epsilon H \geq 4\gamma\sqrt{T}. \tag{22}$$

Call a relevant window a *candidate* if

$$|\mu_I| \geq \frac{\epsilon \bar{h}_t}{2}. \tag{23}$$

This family is deterministic once the law is fixed. On $\mathcal{E}$, if either stream passes either signed support test at I for some $U \in [1,2]$, then I is a candidate, and the passing sign is $\operatorname{sign}(\mu_I)$. To see this, let $\sigma \in \{-1,1\}$ be a passing sign. An interval increment differs from its mean by at most 2η, so

$$\sigma\mu_I \geq \epsilon\bar{h}_t - 2(1+\alpha\epsilon)\eta \geq \frac{\epsilon\bar{h}_t}{2} > 0.$$

Here the actual passing inequality is $\sigma(S_t - S_{t-L}) \geq U\epsilon h_t$, and we have used a conservative bound uniform over $U \in [1,2]$. Thus noncandidate windows cannot cause a switch on $\mathcal{E}$, and at a candidate the opposite sign cannot pass.

For a candidate, (21) and (22) imply

$$\bar{h}_t - \bar{h}_{t-L} \geq \frac{\alpha\epsilon}{4}\bar{h}_t. \tag{24}$$

Set $\kappa = \alpha\epsilon/4 \in (0, 1/4]$. At a fixed scale L, the public L-windows are disjoint. If their candidate endpoint widths in time order are $b_1, \ldots, b_m$, monotonicity and (24) imply $b_j \geq b_{j-1}/(1-\kappa)$, with $b_0 = H$. Consequently

$$\sum_{\substack{I \text{ candidate} \\ \text{at scale } L}} \frac{1}{\bar{h}_{\mathrm{end}(I)}} \leq \frac{1}{H}\sum_{j\geq 1}(1-\kappa)^j \leq \frac{4}{\alpha\epsilon H}. \tag{25}$$

There are J relevant dyadic scales. The endpoint widths and the candidate family in this inequality depend on neither tape nor realized input stream.

We next bound the exceptional shared thresholds. Fix any pair of input streams in $\mathcal{E}$. Because $\mathcal{E}$ is input-only, θ and U retain their independent uniform laws. For a candidate I, put $\sigma = \operatorname{sign}(\mu_I)$ and $\delta_e = 2(1+\alpha\epsilon)\eta$. In either stream, the signed support expression differs from its deterministic reference $\sigma\mu_I - U\epsilon\bar{h}_t$ by at most δ_e. Therefore a difference between the two support decisions requires

$$\left|\sigma\mu_I - U\epsilon\bar{h}_t\right| \leq \delta_e.$$

The possible U values lie in an interval of length at most $2\delta_e/(\epsilon\bar{h}_t)$. Since the support of U has length one, the probability of this event is at most

$$\frac{4(1+\alpha\epsilon)\eta}{\epsilon\bar{h}_t}. \tag{26}$$

No additional sign factor is needed: the preceding argument excluded passage with the opposite sign on $\mathcal{E}$.

The state tests in the two directions compare $S_t - h_t(\theta + U\epsilon)$ and $S_t - h_t(\theta - U\epsilon)$ to zero. Each differs from its corresponding reference with S_t, h_t replaced by $\bar{S}_t, \bar{h}_t$ by at most

$$\delta_s = [1 + \alpha(1+2\epsilon)]\eta.$$

For each fixed U, disagreement on either test is confined to a θ-interval of length at most $2\delta_s/\bar{h}_t$. Uniformity on $[-1,1]$ and independence from U show that the union for the two directions has probability at most

$$\frac{2[1+\alpha(1+2\epsilon)]\eta}{\bar{h}_t}. \tag{27}$$

Include (27) once per candidate window; counting a time more than once only weakens the bound. If no candidate ends at a time, neither stream can switch then. Otherwise, outside the support strips all signed support predicates agree, and outside the state strips both state predicates agree. Starting from the common A_1, induction therefore gives identical played-action paths whenever none of these strip events occurs. This is a pointwise implication. No probability estimate has been conditioned on earlier agreement, on the current action, or on a surviving tape branch.

Applying the union bound and (25) gives, under (22),

$$\begin{aligned}\mathbb{P}\{(A_t^1)_{t=1}^T \neq (A_t^2)_{t=1}^T\} &\le 8Te^{-\eta^2/(2T)} + \left[\frac{4(1+\alpha\epsilon)}{\epsilon} + 2\big(1+\alpha(1+2\epsilon)\big)\right]\frac{4J\eta}{\alpha\epsilon H} \\ &\le 8Te^{-\eta^2/(2T)} + \frac{64J\eta}{\alpha\epsilon^2 H}.\end{aligned} \tag{28}$$

The final inequality uses only $\alpha, \epsilon \le 1$.

With (17), the first term is $r/2$, and (18) bounds the second by $r/2$. The auxiliary conditions also hold: $\gamma\sqrt{T} \le 2\eta$, whereas (18) dominates both $8\eta/\epsilon$ and $4(1+\alpha\epsilon)\eta/\epsilon$. This proves the claimed unconditional whole-path guarantee. □

Remark 6.2. The use of V_t rather than a surplus that vanishes on monotone paths is essential to this argument: every possibly active window forces the growth in (24). The independent U smooths the support comparison itself, while θ smooths the state comparison. All windows remain public and available after switches. The proof makes no assertion that close input paths have identical crossing times without these shared-threshold estimates. For $T = 1$, a common input-independent initialization already replicates exactly.

7 The compensating prediction rule and the mixture

Define on $[-1,1]^T$

$$F(x) = \max\{|S_T(x)|, V_T(x)\} = |S_T(x)| + Q(x). \tag{29}$$

By Lemma 3.1, F is coordinatewise convex and 1-Lipschitz in each coordinate.

Lemma 7.1 (Fair-input value)**.** *For independent uniform signs $\xi_1, \dots, \xi_T$, $\mathbb{E}F(\xi) \le \sqrt{T} + 1$.*

Proof. Hoeffding's inequality gives, for each nonempty interval I,

$$\mathbb{P}\left\{\left|\sum_{i\in I}\xi_i\right| > \gamma\sqrt{|I|}\right\} \le 2e^{-\gamma^2/2}.$$

If $V_T > 0$ at least one interval has this property. There are at most T^2 intervals, so $\mathbb{P}\{V_T > 0\} \le 2T^2e^{-\gamma^2/2} = 1/(8T^2)$. Since $V_T \le T$, its expectation is at most $1/(8T) \le 1$. Finally $\mathbb{E}|S_T| \le \sqrt{\mathbb{E}S_T^2} = \sqrt{T}$ and $F \le |S_T| + V_T$. □

Lemma 7.2 (Fair-future secant prediction)**.** *There is a causal randomized binary-action rule with*

$$\mathbb{E}_R G_T \geq |S_T| + Q - \sqrt{T} - 1 \tag{30}$$

on every deterministic input in $[-1,1]^T$.

Proof. For a fixed prefix $x_{1:t}$ let

$$\Phi_t(x_{1:t}) = \mathbb{E}_{\xi_{t+1:T}} F(x_{1:t}, \xi_{t+1:T}), \qquad \Phi_0 = \mathbb{E}F(\xi).$$

At round t, define the fractional prediction

$$p_t = \frac{\Phi_t(x_{1:t-1}, +1) - \Phi_t(x_{1:t-1}, -1)}{2} \in [-1,1].$$

The range follows from the coordinatewise Lipschitz property. Coordinatewise convexity gives, for the actual $x_t \in [-1,1]$,

$$\Phi_t(x_{1:t}) \leq \frac{1+x_t}{2}\Phi_t(x_{1:t-1}, +1) + \frac{1-x_t}{2}\Phi_t(x_{1:t-1}, -1) = \Phi_{t-1}(x_{1:t-1}) + p_t x_t.$$

Telescoping yields $\sum_t p_t x_t \geq F(x) - \Phi_0$.

The exact conditional expectation need not be computed. At round t, sample a fresh independent Rademacher future $Z^{(t)}_{t+1:T}$, and form

$$\widehat{p}_t = \frac{F(x_{1:t-1}, +1, Z^{(t)}_{t+1:T}) - F(x_{1:t-1}, -1, Z^{(t)}_{t+1:T})}{2} \in [-1,1].$$

Use a fresh independent uniform variable to choose $A_t = +1$ with probability $(1+\widehat{p}_t)/2$. All these variables can be indexed by round on the original tape before inputs arrive. On each fixed deterministic input, $\mathbb{E}_R A_t = p_t$. Thus $\mathbb{E}_R G_T \geq F(x) - \mathbb{E}F(\xi)$, and Lemma 7.1 proves (30). The sampled future is artificial randomness; no actual future cost is accessed. We assert no replication guarantee for this rule alone. □

The complete two-expert algorithm. With the parameters of Section 2, draw one Bernoulli branch coin, independently of every other tape coordinate. With probability β, use the prediction rule of Lemma 7.2 for the entire horizon. Otherwise use the supported relay. Two replicas share this branch coin and all other tape coordinates.

Proof of Theorem 2.1. The parameter choice has $\lambda \in (0,1)$. Lemma 5.8 and the first two terms of (6) give relay reward at least $|S_T| - \lambda Q - 20H$ in original-tape expectation. By Lemma 7.2, the full mixture therefore has

$$\begin{aligned}\mathbb{E}_R G_T &\geq (1-\beta)(|S_T| - \lambda Q - 20H) + \beta(|S_T| + Q - \sqrt{T} - 1) \\ &= |S_T| - 20(1-\beta)H - \beta(\sqrt{T}+1),\end{aligned}$$

because $(1-\beta)\lambda = \beta$. Equation (2) gives $\mathrm{Reg}_A \leq 10H + (\beta/2)(\sqrt{T}+1) = B(T,\rho)$. No branch or negative-regret contribution has been discarded.

For every fixed product law, Lemma 6.1 and the third term in (6) bound the relay's entire-path failure by r. Charge the compensating branch its full probability β, whether or not its two paths happen to agree. Independence of the branch coin from all inner randomness yields failure at most

$$\beta + (1-\beta)r = \rho/4 + (1-\rho/4)\rho/2 < \rho.$$

The arbitrary cost-vector reduction was established in (2). For $T = 1$, a shared uniform expert is independent of both costs, replicates exactly, and has expected regret at most $1/2$. □

8 Extension to any fixed number of experts

The two-expert learner can be composed in a tournament. Its actual child loss sequences are adaptive functions of the original observations and need not be product-distributed. We instead introduce deterministic modal paths for the analysis of replication, conditional only on tapes.

Lemma 8.1 (A modal whole-path comparison)**.** *Let $Z = f(X, R)$ take values in a finite alphabet, where the input X has a fixed law and the original tape R is independent of X. For each tape value s, define $p_s(z) = \mathbb{P}_X\{f(X, s) = z\}$ and select a maximizer $z^*(s)$ with a fixed lexicographic rule for ties. If X, X' are independent copies sharing R, then*

$$\mathbb{P}_{R,X}\{f(X, R) \neq z^*(R)\} \leq \mathbb{P}_{R,X,X'}\{f(X, R) \neq f(X', R)\}. \tag{31}$$

Proof. The probabilities $p_s(z)$ are measurable functions of s, by integration of the measurable indicators defining f. There are finitely many z, so lexicographic maximization gives a measurable selection. For each fixed s, independence of X, X' gives conditional disagreement $1 - \sum_z p_s(z)^2$. Since

$$\sum_z p_s(z)^2 \leq \max_z p_s(z),$$

the single-input modal deviation $1 - \max_z p_s(z)$ is no larger. Integrating over the original tape proves (31). □

The modal path may depend on the unknown input law. It is used only in the analysis and is neither computed nor supplied to the learner. In particular, (31) is an averaged-tape inequality, not a bound on deviation for every fixed tape.

Theorem 8.2 (Tournament reduction)**.** *Suppose a two-expert learner, for every $r \in (0, 1)$, has expected regret at most $B(T, r)$ on every deterministic sequence of costs in $[0, 1]^2$ and is r-replicable for every fixed time-varying product law on those cost pairs. The expectation and replication probability use its entire original input-independent tape.*

For $n \geq 2$, $\rho \in (0, 1)$, and $T \geq 2$, let

$$N = 2^{\lceil \log_2 n \rceil}, \qquad D_{\mathrm{tr}} = \log_2 N, \qquad r = \frac{\rho}{N^2}.$$

There is an n-expert learner with expected regret at most $D_{\mathrm{tr}} B(T, r)$ on every deterministic cost sequence, whose entire original-expert action path has disagreement probability at most

$$\frac{r(N^2 - 1)}{3} < \rho \tag{32}$$

under any fixed nonstationary product law on $[0, 1]^{n \times T}$. Coordinates within a round may be dependent.

Proof. Pad the original experts with duplicates to form N leaves of a complete balanced binary tree. Each leaf recommends its fixed original expert index. Each internal node runs a copy of the two-expert learner with parameter r on its own independent subtape. At a round, a node selects one of its two children and recommends the original index currently recommended by that child. The root recommendation is played.

After the full original cost vector is revealed, every internal node receives the costs of both current child recommendations and updates. All nodes operate in full information at every round, including nodes not selected by their ancestors. Consequently a child's recommendation path depends only on past original costs and tapes in that child's subtree; it depends on neither its parent's tape nor any ancestor's choices. This also specifies a causal implementation: all current recommendations are determined before the current cost vector is revealed.

Regret. Fix a deterministic original cost sequence. Conditional on all tapes strictly below an internal node, its two child recommendation paths and their two cost sequences are deterministic and independent of its own tape. The node's expected total cost is therefore at most the smaller child total plus $B(T, r)$, and hence at most the total of either particular child plus $B(T, r)$. Integrating the descendant tapes gives

$$\mathbb{E}L_{\text{node}} \le \mathbb{E}L_{\text{child}} + B(T, r)$$

for either specified child. Iterating along the path to any fixed leaf yields

$$\mathbb{E}L_{\text{root}} \le L_{\text{leaf}} + D_{\text{tr}}B(T, r).$$

Choosing a leaf representing a best original expert proves the regret bound. This argument keeps all tape averaging intact and makes no assumption that any child regret is nonnegative.

Replication. Fix an arbitrary product law on the original cost vectors. For a parent, let q_L, q_R be bounds on the disagreement probabilities of its child original-index paths under that law, using independent input streams and the same subtree tapes. Condition only on all tapes strictly below the parent. For each child, choose its canonical modal whole original-index path under the fixed law and those fixed tapes, as in Lemma 8.1. These two paths are now deterministic sequences of indices. Their conditional definition may use the entire fixed law; it is an analysis device, not part of the causal algorithm.

Construct a shadow parent in each stream by evaluating the current original cost vector at these two deterministic shadow indices and running the parent learner on the resulting cost pairs with its original parent tape. For every fixed choice of descendant tapes, the shadow cost pairs are independent across time, since the original vectors are. The two shadow streams are independent copies of the same product law. The parent's own tape remains independent of these inputs and of the fixed descendant tapes. Its two-expert guarantee therefore bounds the shadow parent decision-path disagreement by r. Integrating over the descendant tapes preserves this bound.

Lemma 8.1, followed by a union bound over both streams and both children, bounds the unconditional probability that any actual child path differs from its shadow by

$$2(q_L + q_R).$$

Only the averaged-tape shadow inequality is used here. We do not condition on the complement of this deviation event, and we do not assert that the actual adaptive child costs form a product law.

On the event of no child-shadow deviation, the actual and shadow parent cost sequences coincide within each stream. Causality and use of the same parent tape then imply that their parent decision paths coincide within that stream. If the two shadow parent decision paths also agree, the selected deterministic shadow child indices agree at every time, so the actual parent original-index paths agree. This pointwise event containment proves

$$q_{\text{parent}} \le r + 2(q_L + q_R). \tag{33}$$

No independence between left and right child paths is required.

Leaves have zero disagreement. In a complete balanced tree, (33) gives, at height j, the bound $q_j \le r + 4q_{j-1}$, with $q_0 = 0$. Hence

$$q_{\text{root}} \le r \sum_{j=0}^{D_{\text{tr}}-1} 4^j = \frac{r(N^2-1)}{3} < \rho,$$

as claimed. All probabilities include the entire original collection of node tapes, including any rare branch coins used by the base learners. □

Applying Theorem 8.2 to the two-expert bound already proved gives

$$\mathrm{Reg}_T = O_{n,\rho}\Big(\sqrt{T}\log^{7/2}(2T)\Big)$$

for every fixed finite n and fixed $\rho \in (0,1)$. The dependence on n and ρ is poor; the reduction makes no claim of an optimal dependence on either parameter. For $n = 1$, always select the sole expert. For $T = 1$, any shared input-independent expert choice replicates exactly and has regret at most one. These conventions cover the finite boundary cases without the tournament parameters.

9 Computation and scope

The interval functional admits the dynamic program

$$V_t = \max\left\{V_{t-1}, \max_{0\le s<t}\left[V_s + (|S_t - S_s| - \gamma\sqrt{t-s})_+\right]\right\}. \tag{34}$$

To verify this identity, take the last interval of an optimal collection. If it ends before t, its value is at most V_{t-1}; otherwise its predecessors lie in the prefix ending at its left endpoint. Conversely every candidate in (34) is feasible. Prefix sums and this dynamic program require $O(T^2)$ arithmetic operations. The relay's public window tests require at most $O(T\log(2T))$ additional operations. Evaluating F twice for each fresh artificial future gives a naive $O(T^3)$ arithmetic implementation of the compensating rule. A tournament with $O(n)$ internal nodes has the corresponding $O(nT^3)$ arithmetic bound.

These are exact real-arithmetic descriptions. No finite-precision comparison rule, bit-complexity bound, implementation measurement or practical speedup follows. The explicit constants and dependence on n, ρ can make the numerical bound uninformative at moderate horizons. The asymptotic improvement concerns T with those parameters fixed. The algorithm knows T and ρ; an anytime guarantee is not claimed.

The rare branch is allowed because regret is an unconditional expectation over the entire input-independent tape, whereas replication permits a probability ρ of failure. The proof does not provide a high-probability or seedwise regret theorem. It also does not extend the product-law replication guarantee to input laws that adapt to realized past costs or actions. The logarithmic factors leave the strict square-root alternative in [1] unresolved by this work.

References

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